CLARIO COACH

AI-Powered Customer Support Coaching Assistant

PROJECT DOCUMENTATION

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1. Abstract

Clario Coach is an AI-powered customer support coaching assistant designed to help support agents improve communication quality, consistency, and efficiency. It analyzes customer messages for intent and sentiment, retrieves relevant information from a domain-specific knowledge base, and uses Google Gemini to generate context-aware coaching and response suggestions. The platform combines NLP, semantic search, vector embeddings, Retrieval-Augmented Generation (RAG), React, FastAPI, and ChromaDB.

2. Introduction

Customer-support agents often need to understand customer intent, recognize emotional tone, locate relevant information, and formulate accurate responses quickly. Clario Coach addresses this challenge through AI-assisted guidance during simulated support conversations. The system combines conversation analysis with knowledge retrieval so generated suggestions can be grounded in relevant information.

3. Problem Statement

Conventional support workflows depend heavily on manual knowledge-base searches, predefined responses, and individual agent experience. This can increase response time and produce inconsistent communication. Keyword-only search can also fail when the wording of a customer query differs from wording stored in the knowledge base.

4. Objectives

• Assist support agents with context-aware response suggestions.
• Identify customer intent and sentiment.
• Retrieve relevant information using semantic search.
• Use RAG to provide relevant context to the language model.
• Improve consistency and efficiency in customer communication.
• Provide an integrated web interface and API-based backend.

5. Existing System

Traditional support environments often require agents to manually search documentation and construct responses. Separate tools may be used for sentiment analysis, knowledge retrieval, and response drafting. These disconnected workflows can increase cognitive load, response time, and inconsistency. General-purpose AI may also generate responses without access to organization-specific knowledge.

6. Proposed System

Clario Coach integrates customer-message analysis, knowledge retrieval, and AI response generation into a single workflow. Intent and sentiment are analyzed, relevant knowledge is retrieved through vector similarity, and retrieved context is supplied to Google Gemini to generate coaching and response suggestions. The React frontend communicates with FastAPI services through APIs.

7. System Architecture & Workflow

Customer Message → Intent Detection → Sentiment Analysis → Knowledge Retrieval (RAG) → Google Gemini → Coaching / Response Suggestion → Response Display. The knowledge workflow processes documents into searchable vector representations before they are used during conversations.

8. Major Modules

User Interface: Support-agent workspace and conversation experience.
Customer Conversation: Simulated customer interactions and session data.
Intent Detection: Identifies the likely purpose of a customer message.
Sentiment Analysis: Determines emotional tone.
Knowledge Management: Document upload and indexing.
Semantic Search: Vector-based information retrieval.
RAG: Combines retrieved context with the current query.
AI Coaching: Generates response recommendations using Gemini.
Session Management: Starts and retrieves coaching sessions.

9. RAG Pipeline

1. Upload knowledge-base documents.
2. Extract and preprocess text.
3. Split text into chunks.
4. Generate embeddings using Sentence Transformers.
5. Store embeddings in ChromaDB.
6. Convert the incoming query into an embedding.
7. Perform semantic similarity search.
8. Retrieve relevant context.
9. Supply context to Gemini.
10. Generate a context-aware coaching response.

10. Technology Stack

Frontend: React, TypeScript, Vite, Tailwind CSS.
Backend: Python, FastAPI, Uvicorn, Pydantic.
AI / NLP: Google Gemini, Sentence Transformers, all-MiniLM-L6-v2, semantic search, RAG.
Vector Database: ChromaDB.
Development: Git, GitHub, REST APIs, Swagger / OpenAPI.

11. Project Structure

clarion-coach/ → backend/app (agents, api, core, rag, schemas, main.py), backend/requirements.txt, public/, src/, .env.example, .gitignore, LICENSE, package.json, README.md.

12. API Endpoints

GET / — Health check.
GET /docs — Swagger documentation.
POST /chat — AI coaching interaction.
POST /upload — Knowledge-base document upload.
POST /knowledge/search — Semantic knowledge search.
GET /session/latest — Retrieve latest session.
POST /session/start — Start a coaching session.

13. Installation & Execution

Backend: Navigate to backend, create a Python 3.11 virtual environment, activate it, install dependencies with pip install -r requirements.txt, configure required environment variables locally, and run uvicorn app.main:app --reload.

Frontend: From the project root, run npm install and npm run dev. Access the local URL displayed by Vite.

14. Security & Configuration

Sensitive values such as API keys and credentials must be stored in environment variables and must never be committed to a public repository. The actual .env file should remain local and be excluded through .gitignore. The repository should contain safe placeholder configuration such as .env.example.

15. Testing & Validation

Validation includes frontend loading, backend health checks, API availability through Swagger, session creation, customer-message processing, intent and sentiment analysis, knowledge retrieval, and AI-generated coaching responses. Multiple simulated customer conversations can be used to verify contextual behavior.

16. Advantages

• Integrated support analysis and AI coaching.
• RAG-based context-aware generation.
• Semantic retrieval instead of only keyword matching.
• Interactive web interface.
• Documented REST APIs.
• Extensible knowledge-retrieval architecture.

17. Applications

Customer-support training, simulated support environments, agent assistance, knowledge-base navigation, response-quality improvement, and AI-assisted communication coaching.

18. Limitations

The current implementation uses simulated customer interactions and depends on configured AI and embedding services. AI-generated suggestions should be reviewed by a human agent before use in real customer communication. Production deployment would require stronger authentication, privacy controls, monitoring, and access management.

19. Future Enhancements

Voice-based coaching; live-chat integration; agent-performance dashboards; analytics and reporting; multilingual support; richer conversation history; authentication and role-based access control; personalized coaching; stronger guardrails; enhanced monitoring and evaluation.

20. Conclusion

Clario Coach demonstrates how modern AI techniques can be integrated into a customer-support coaching platform. By combining intent and sentiment analysis, semantic search, ChromaDB vector retrieval, RAG, and Google Gemini, the system provides context-aware assistance to support agents.

21. License

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22. Author

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